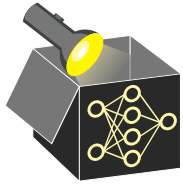
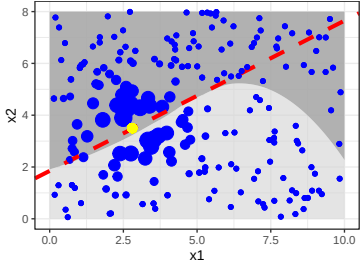
Interpretable Machine Learning



Local Explanations: LIME

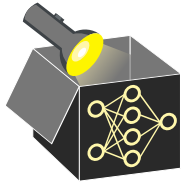
Local Interpretable Model-agnostic Explanations (LIME)



Learning goals

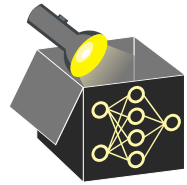
- Understand motivation for LIME
- Develop a mathematical intuition

LIME



- **Locality assumption:**
  $\hat{f}$ behaves similarly simple in a small neighborhood of $\boldsymbol{x}$
  $\rightsquigarrow$ Approximate $\hat{f}$ near $\boldsymbol{x}$ using an interpretable surrogate model $\hat{g}$
- **Interpretation strategy:**
  Use $\hat{g}$'s simple internal structure to explain $\hat{f}(\boldsymbol{x})$ locally
  $\rightsquigarrow$ **Common surrogates:** sparse linear models, shallow decision trees
- **Applicability:** Model-agnostic; supports tabular, image, and text data
- **In practice:** Generate samples near $\boldsymbol{x}$, predict with $\hat{f}$, and fit $\hat{g}$ to these samples using $\hat{f}$'s outputs as targets, weighting samples by their proximity to $\boldsymbol{x}$

LIME: CHARACTERISTICS



Definition:
LIME provides a local explanation for a black-box model $\hat{f}$ in form of a surrogate model $\hat{g} \in \mathcal{G}$, where $\mathcal{G}$ is a class of interpretable models

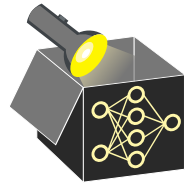
Surrogate model $\hat{g}$ should satisfy two characteristics:

1. **Interpretable**: Provide human-understandable insights into the relationship between input features and prediction, e.g. via coefficients or model structure
2. **Local fidelity** / **faithfulness**: $\hat{g}$ closely approximates $\hat{f}$ in the vicinity of the input $\boldsymbol{x}$ being explained

Goal: Find $\hat{g}$ with **minimal complexity and maximal local fidelity**

MODEL COMPLEXITY

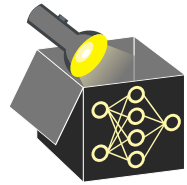
We can measure complexity of $\hat{g} \in \mathcal{G}$ using a complexity measure $J : \mathcal{G} \to \mathbb{R}_0$



- **Example: (Sparse) Linear Models**
  Let $\mathcal{G} = \{g : \mathcal{X} \to \mathbb{R} \mid g(\boldsymbol{x}) = s(\boldsymbol{\theta}^\top \boldsymbol{x})\}$ be the class of linear models
  - $s(\cdot)$ is the identity (linear model) or logistic sigmoid function (logistic regression)

  $\rightsquigarrow$ $J(g) = \sum_{j=1}^{p} \mathbb{1}_{\{\theta_j \neq 0\}}$: count the number of non-zero coefficients via the L_0 norm of $\boldsymbol{\theta}$

MODEL COMPLEXITY



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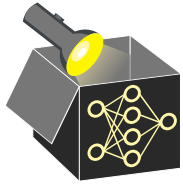
  $\rightsquigarrow$ $J(g) = \sum_{j=1}^{p} \mathbb{1}_{\{\theta_j \neq 0\}}$: count the number of non-zero coefficients via the L_0 norm of $\boldsymbol{\theta}$
- **Example: Decision Trees**
  Let $\mathcal{G} = \{g : \mathcal{X} \to \mathbb{R} \mid g(\boldsymbol{x}) = \sum_{m=1}^{M} c_m \mathbb{1}_{\{\boldsymbol{x} \in \mathcal{R}_m\}}\}$ be the class of trees
  - $\mathcal{R}_m$ are disjoint axis-parallel regions (leaves); $c_m \in \mathbb{R}$ are constant predictions

  $\rightsquigarrow$ $J(g) = M$: count the number of terminal or leaf nodes

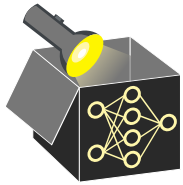
LOCAL FIDELITY OF SURROGATE MODELS

- Surrogate $\hat{g}$ is **locally faithful** to a black-box model $\hat{f}$ around an input $\boldsymbol{x}$ if

$$\hat{g}(\mathbf{z}) \approx \hat{f}(\mathbf{z}) \quad \text{for synthetic samples } \mathbf{z} \in \mathcal{Z} \subseteq \mathbb{R}^p \text{ generated around } \boldsymbol{x}$$



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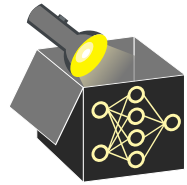


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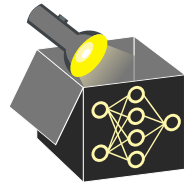
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- To operationalize this optimization, we need:
  1. **A proximity (similarity) measure** $\phi_{\boldsymbol{x}}(\mathbf{z})$ between $\mathbf{z}$ and $\boldsymbol{x}$, e.g.:

$$\phi_{\boldsymbol{x}}(\mathbf{z}) = \exp(-d(\boldsymbol{x}, \mathbf{z})^2 / \sigma^2)$$

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  - d is a distance metric, e.g. Euclidean or Gower for mixed types
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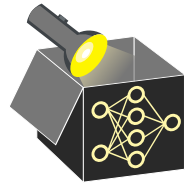
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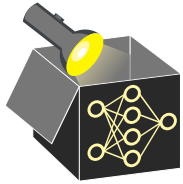
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- The overall **local fidelity objective** is measured by a weighted loss:

$$L(\hat{f}, \hat{g}, \phi_{\boldsymbol{x}}) = \sum_{\mathbf{z} \in \mathcal{Z}} \phi_{\boldsymbol{x}}(\mathbf{z}) \cdot L(\hat{f}(\mathbf{z}), \hat{g}(\mathbf{z}))$$

LIME OPTIMIZATION TASK



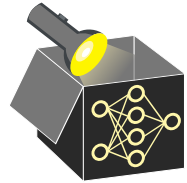
- Optimization problem of LIME:

$$\arg\min_{\hat{g}\in\mathcal{G}} L(\hat{f}, \hat{g}, \phi_{\boldsymbol{x}}) + J(\hat{g})$$

- **In practice** LIME uses a two-stage approach:
  1. User sets complexity $J(\hat{g})$ beforehand, e.g. LASSO with k features
  2. Optimize $L(\hat{f}, \hat{g}, \phi_{\boldsymbol{x}})$ for fixed complexity
- **Goal:** Build a **model-agnostic** explainer
  - $\rightsquigarrow$ Optimize $L(\hat{f}, \hat{g}, \phi_{\boldsymbol{x}})$ without making assumptions on the form of $\hat{f}$
  - $\rightsquigarrow$ Surrogate $\hat{g}$ approximates $\hat{f}$ locally through sampling and fitting

LIME ALGORITHM: OUTLINE

▸ "Ribeiro." 2016

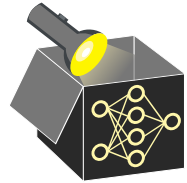


Input:

- Pre-trained black-box model $\hat{f}$
- Observation $\boldsymbol{x}$ whose prediction $\hat{f}(\boldsymbol{x})$ we want to explain
- Interpretable model class $\mathcal{G}$ for the local surrogate

LIME ALGORITHM: OUTLINE

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Input:

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- Observation $\boldsymbol{x}$ whose prediction $\hat{f}(\boldsymbol{x})$ we want to explain
- Interpretable model class $\mathcal{G}$ for the local surrogate

Algorithm:

1. Independently sample new points $\mathbf{z} \in \mathcal{Z}$
2. Retrieve predictions $\hat{f}(\mathbf{z})$ for the obtained points $\mathbf{z}$
3. Weight $\mathbf{z} \in \mathcal{Z}$ by their proximity $\phi_{\boldsymbol{x}}(\mathbf{z})$ to quantify closeness to $\boldsymbol{x}$
4. Train interpretable surrogate model $\hat{g}$ on data $\mathbf{z} \in \mathcal{Z}$ using weights $\phi_{\boldsymbol{x}}(\mathbf{z})$
   $\rightsquigarrow$ Predictions $\hat{f}(\mathbf{z})$ are used as the target of this model
5. Return $\hat{g}$ as the local explanation for $\hat{f}(\boldsymbol{x})$

LIME ALGORITHM: EXAMPLE

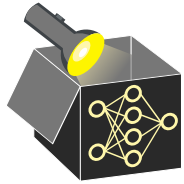
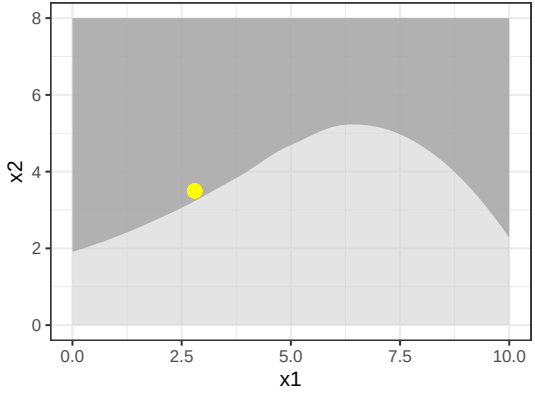
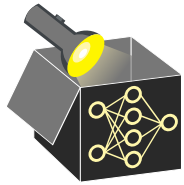


Illustration of LIME based on a classification task:

- Light/dark gray background: prediction surface of a classifier
- Yellow point: $\boldsymbol{x}$ to be explained
- $\mathcal{G}$: class of logistic regression models



LIME ALGO.: EXAMPLE (STEP 1+2: SAMPLING)



Strategies for sampling:

- Uniformly sample new points from the feasible feature range
- Use the training data set with or without perturbations
- Draw samples from the estimated univariate distribution of each feature
- Create an equidistant grid over the supported feature range

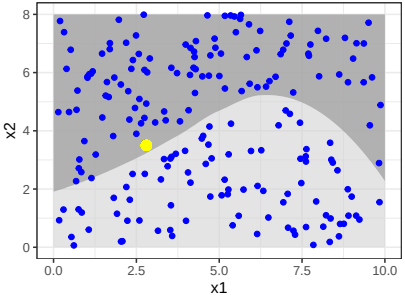


Figure: Uniformly sampled

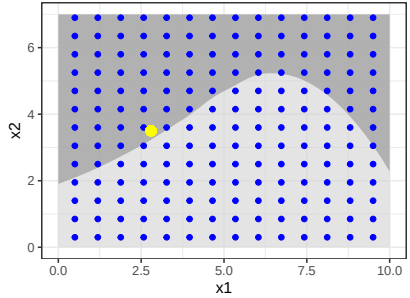
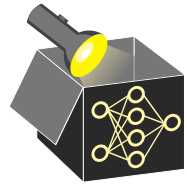


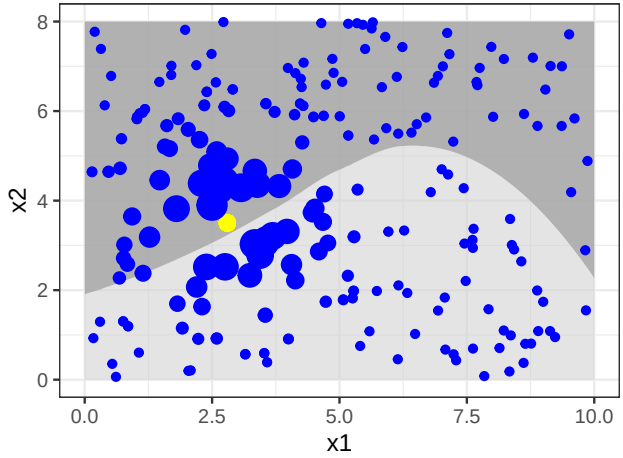
Figure: Equidistant grid

LIME ALGO.: EXAMPLE (STEP 3: PROXIMITY)



In this example, we use the exponential kernel defined on the Euclidean distance d:

$$\phi_{\boldsymbol{x}}(\mathbf{z}) = \exp(-d(\boldsymbol{x}, \mathbf{z})^2 / \sigma^2)$$



LIME ALGO.: EXAMPLE (STEP 4: SURROGATE)

In this example, we fit a **logistic regression** model
$\rightsquigarrow L(\hat{f}(\mathbf{z}), \hat{g}(\mathbf{z}))$ is the Bernoulli loss

